

B.Tech. Degree II Semester Regular Examination July 2024

CE/ME/SE 23-200-0205A BASIC ELECTRONICS ENGINEERING
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes
On successful completion of the course, the students will be able to:

- CO1: Illustrate the concept of diode in rectifiers, filter circuits and wave shaping.
CO2: Interpret the functioning of oscillators and operational amplifiers.
CO3: Explain the principle of embedded systems and sensors.
CO4: Summarise the functioning of a communication system and different modulation technologies.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a)	What is meant by doping? How are p-type semiconductors formed?	2	L2	1	1
(b)	An operational amplifier with a negative feedback applied produces an output of 2.5 V when an input of 5.5 mV is present. Determine the closed-loop voltage gain.	2	L5	2	1,5
(c)	Differentiate RISC and CISC processors.	2	L1	3	1
(d)	Differentiate HLR and VLR in GSM architecture of mobile communication.	2	L2	4	1
(e)	List out the reasons for, choosing the downlink frequency lower than that of the uplink frequency in satellite communication.	2	L2	4	

PART B
(4 × 10 = 40)

II.	(a)	Explain the working of a full wave bridge rectifier using filter with circuit diagram and relevant input, output waveforms.	7	L3	1	1.
	(b)	An amplifier produces an output voltage of 8 V for an input of 100 mV. If the input and output currents in this conditions are 8 mA and 200 mA respectively, determine the voltage gain, current gain and power gain of the amplifier.	3	L3	1	1
OR						
III.	(a)	What do you mean by multistage amplifiers? Explain the working of a two stage RC coupled amplifier with a neat circuit diagram.	6	L3	1	
	(b)	Discuss the need of voltage regulators in power supply systems with necessary circuit diagram.	4	L3	1	
IV.	(a)	Explain the inverting and non-inverting configurations of an op-amp. List out the ideal op-amp characteristics.	7	L2	2	
	(b)	Implement the EX-NOR function using basic gates.	3	L3	2	
OR						
V.	(a)	Explain the working of a Ladder network oscillator using BJT with a neat circuit diagram.	7	L3	2	
	(b)	Define CMRR and slew rate of an op-amp with typical values.	3	L2	2	

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		Marks	BL	CO	PO
VI.	(a) With a block diagram briefly describe the elements of an embedded system and explain about the core in detail.	7	L2	3	1,5,12
	(b) What are the differences between a microprocessor and a microcontroller?	3	L2	3	1,5,12
OR					
VII.	(a) Explain the working of a 7-segment LED display with suitable diagrams and examples. Discuss the two types of configurations with necessary diagrams.	7	L2	3	1,5
	(b) What are transducers? Differentiate sensors and actuators used in an embedded system with suitable examples.	3	L2	3	1
VIII.	(a) Explain in detail the working of an optical communication system with a neat schematic diagram. Mention any two advantages and disadvantages.	7	L4	4	1,5
	(b) What do you mean by multiplexing? Differentiate TDM and FDM with suitable examples.	3	L2	4	1,5,12
OR					
IX.	(a) Define modulation and briefly explain AM and FM with necessary diagrams.	5	L2	4	1,5,12
	(b) Define satellite communication and briefly explain the working of earth station transmitter employed in the satellite communication with necessary block diagrams.	5	L3	4	1,5,12

Bloom's Taxonomy Levels

L1 – 5%, L2 – 50%, L3 – 30%, L4 – 5%, L5 – 10%.
